

Chemistry

SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
 Full Marks : +3 If **ONLY** the correct option is chosen;
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
 Negative Marks : -1 In all other cases.

***Q.1** A closed vessel contains 10 g of an ideal gas **X** at 300 K, which exerts 2 atm pressure. At the same temperature, 80 g of another ideal gas **Y** is added to it and the pressure becomes 6 atm. The ratio of root mean square velocities of **X** and **Y** at 300 K is

- (A) $2\sqrt{2} : \sqrt{3}$ (B) $2\sqrt{2} : 1$
 (C) $1 : 2$ (D) $2 : 1$

Ans. (D)

Sol. $2 \times V = \frac{10}{M_x} \times R \times T \quad \dots(1)$

$4 \times V = \frac{80}{M_y} \times R \times T \quad \dots(2)$

$2 = \frac{80 \times M_x}{M_y \times 10}$

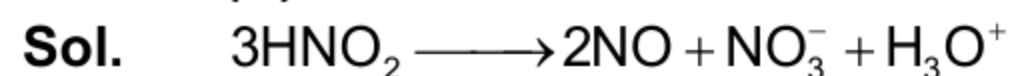
$\frac{M_x}{M_y} = \frac{1}{4}$

$\frac{(V_{rms})_x}{(V_{rms})_y} = \sqrt{\frac{4}{1}} = \frac{2}{1}$

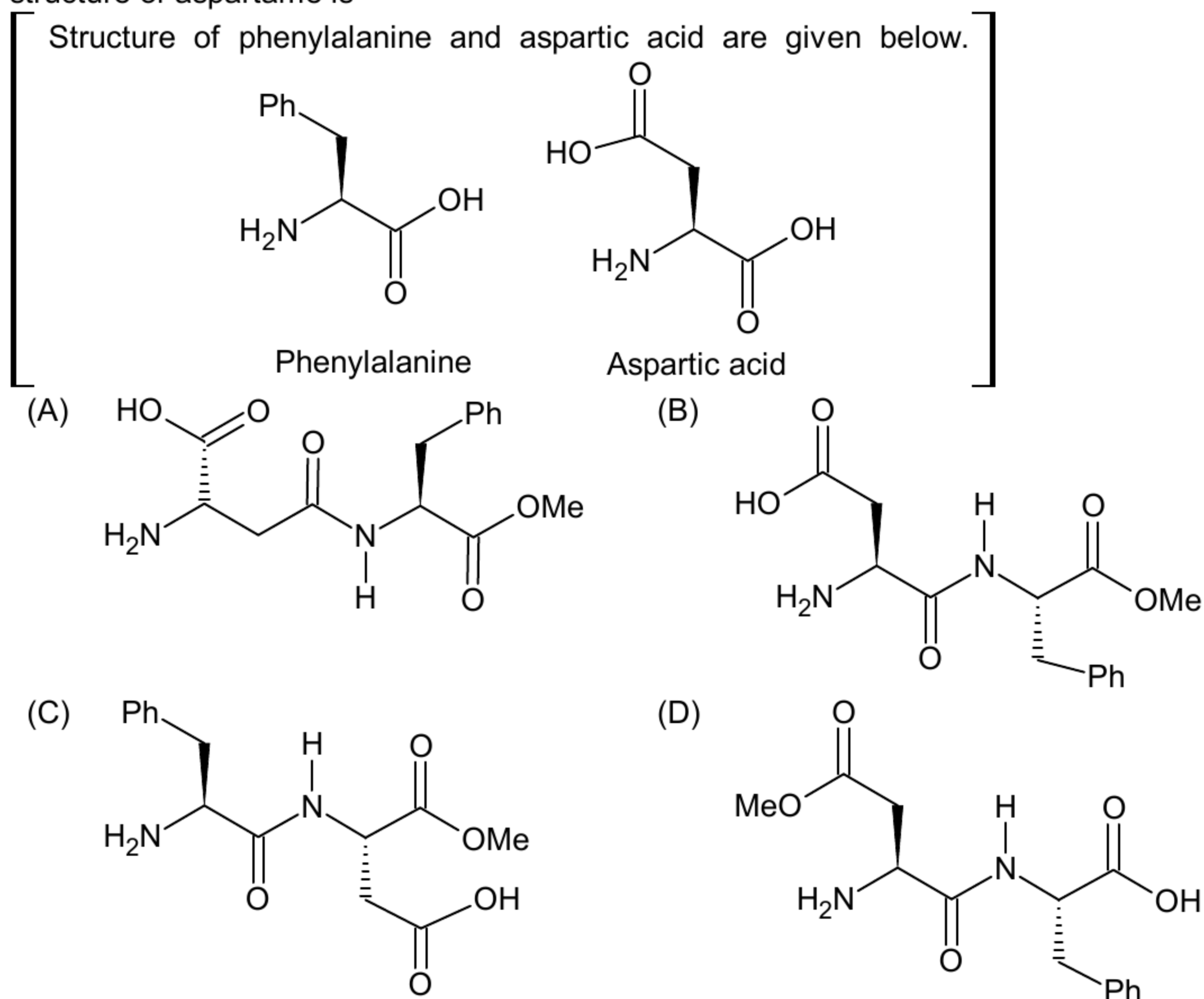
Q.2 At room temperature, disproportionation of an aqueous solution of *in situ* generated nitrous acid (HNO_2) gives the species

- (A) H_3O^+ , NO_3^- and NO (B) H_3O^+ , NO_3^- and NO_2
 (C) H_3O^+ , NO^- and NO_2 (D) H_3O^+ , NO_3^- and N_2O

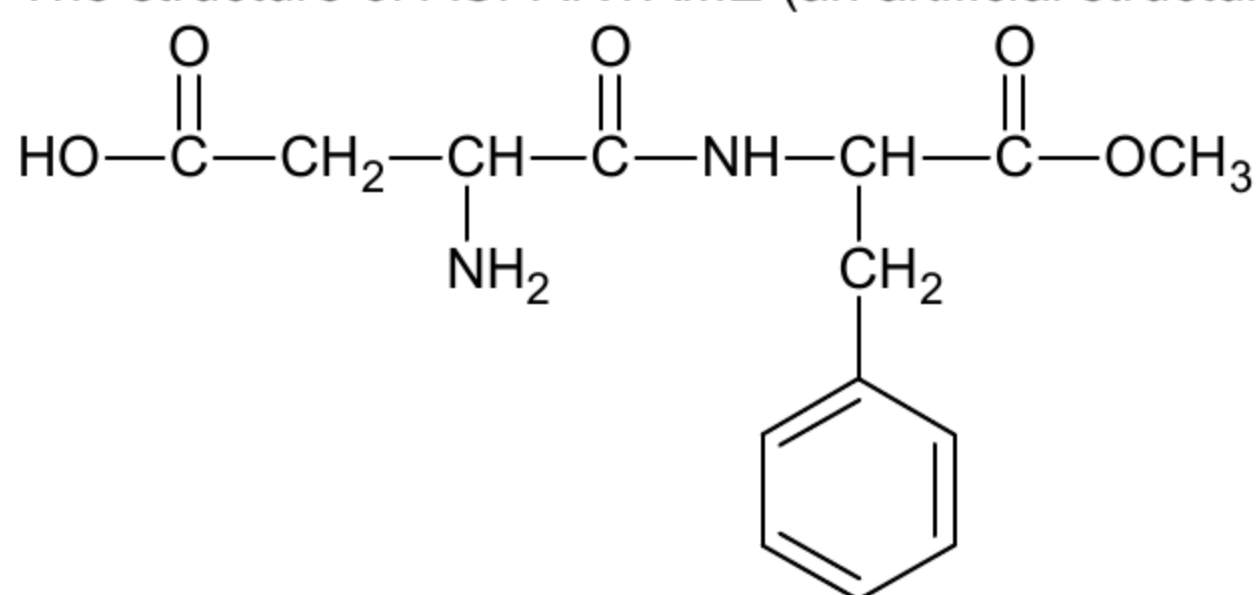
Ans. (A)



Q.3 Aspartame, an artificial sweetener, is a dipeptide aspartyl phenylalanine methyl ester. The structure of aspartame is



Ans. (B)
Sol. The structure of ASPARTAME (an artificial structure) is:

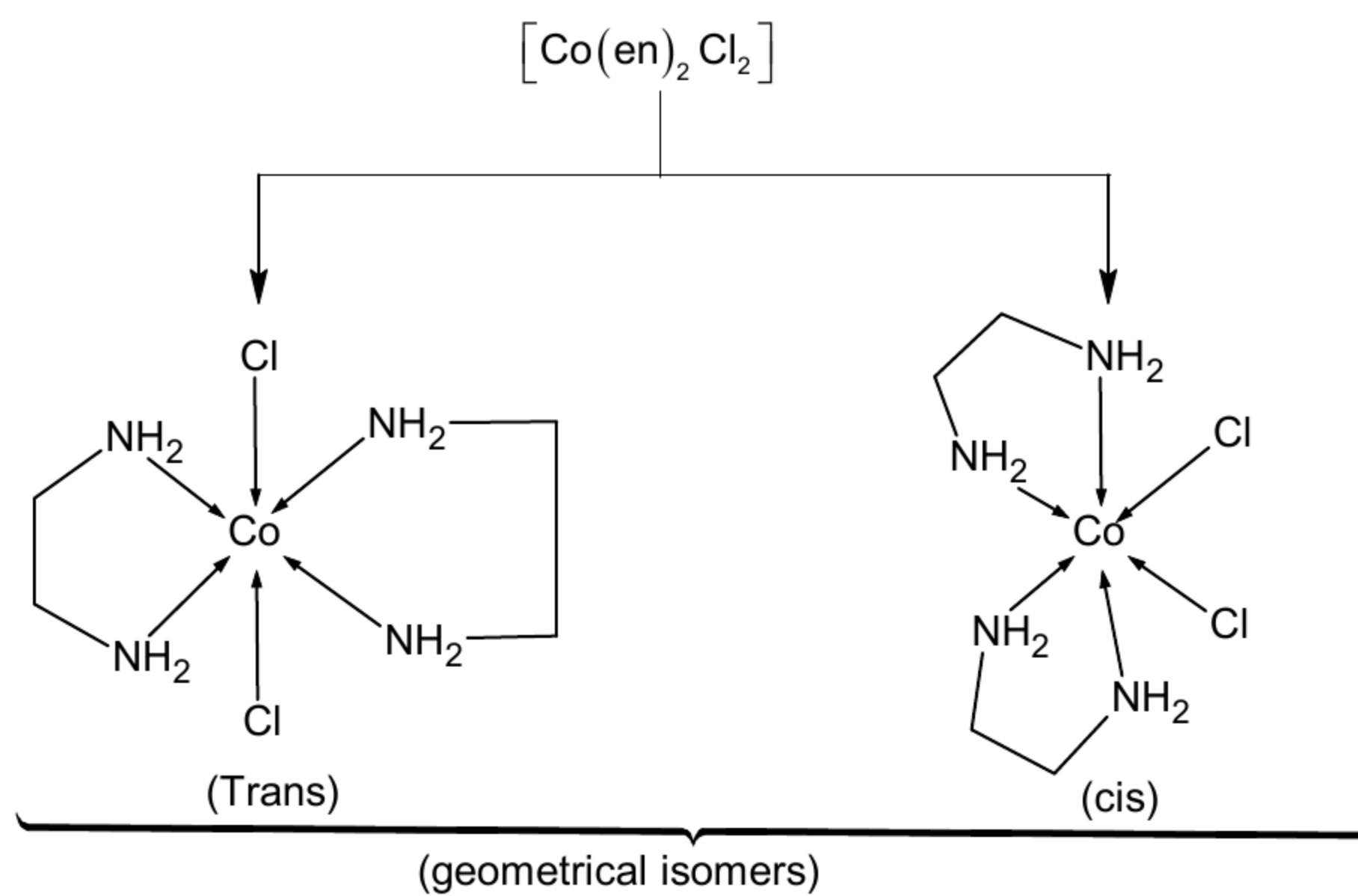
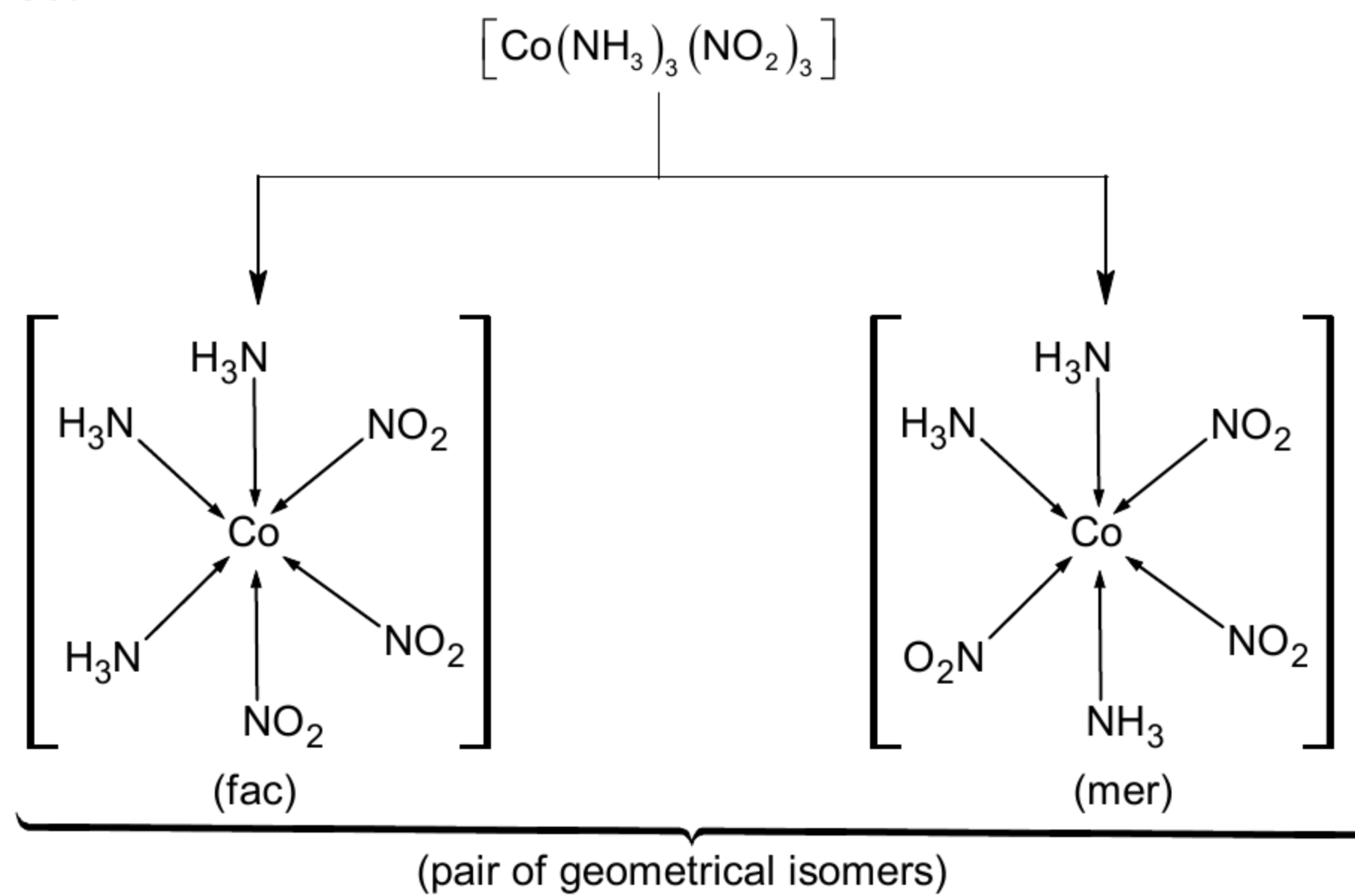


Q.4 Among the following options, select the option in which each complex in **Set-I** shows geometrical isomerism and the two complexes in **Set-II** are ionization isomers of each other.

[en = H₂NCH₂CH₂NH₂]

- (A) **Set – I:** [Ni(CO)₄] and [PdCl₂(PPh₃)₂]
Set – II: [Co(NH₃)₅Cl]SO₄ and [Co(NH₃)₅(SO₄)]Cl
- (B) **Set – I:** [Co(en)(NH₃)₂Cl₂] and [PdCl₂(PPh₃)₂]
Set – II: [Co(NH₃)₆][Cr(CN)₆] and [Cr(NH₃)₆][Co(CN)₆]
- (C) **Set – I:** [Co(NH₃)₃(NO₂)₃] and [Co(en)₂Cl₂]
Set – II: [Co(NH₃)₅Cl]SO₄ and [Co(NH₃)₅(SO₄)]Cl
- (D) **Set – I:** [Cr(NH₃)₅Cl]Cl₂ and [Co(en)(NH₃)₂Cl₂]
Set – II: [Cr(H₂O)₆]Cl₃ and [Cr(H₂O)₅Cl]Cl₂·H₂O

Ans. (C)
Sol. Set – I



$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Cl}$ are ionization isomers.

SECTION 2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
 - Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen;
 - Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
 - Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
 - Zero Marks* : 0 If unanswered;
 - Negative Marks* : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 - choosing **ONLY** (A), (B) and (D) will get +4 marks;
 - choosing **ONLY** (A) and (B) will get +2 marks;
 - choosing **ONLY** (A) and (D) will get +2 marks;
 - choosing **ONLY** (B) and (D) will get +2 marks;
 - choosing **ONLY** (A) will get +1 mark;
 - choosing **ONLY** (B) will get +1 mark;
 - choosing **ONLY** (D) will get +1 mark;
 - choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 - choosing any other option(s) will get -2 marks.

- *Q5.** Among the following, the correct statement(s) for electrons in an atom is(are)
- (A) Uncertainty principle rules out the existence of definite paths for electrons.
 - (B) The energy of an electrons in 2s orbital of an atom is lower than the energy of an electron that is infinitely far away from the nucleus.
 - (C) According to Bohr's model, the most negative energy value for an electron is given by $n = 1$, which corresponds to the most stable orbit.
 - (D) According to Bohr's model, the magnitude of velocity of electrons increases with increase in values of n .

Ans. (A, B, C)

Sol. (A) uncertainty principle rules – out the existence of definite paths for an electron due to wave nature of electron.

$$(B) \text{ Energy} \propto -\frac{Z^2}{n^2}$$

$$\text{So, } E_{2s} < E_{\infty}$$

$$(C) E = -\frac{13.6 \times Z^2}{n^2}$$

$$\text{If } n = 1$$

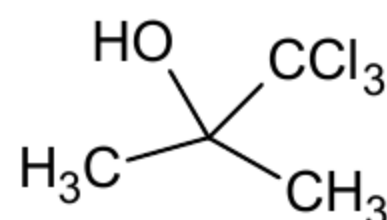
$$E = -(13.6 \times Z^2) \text{ eV is most negative value and therefore most stable.}$$

$$(D) V = 2.186 \times 10^6 \times \frac{Z}{n}$$

So, (D) is wrong.

- Q.6** Reaction of *iso*-propylbenzene with O_2 followed by the treatment with H_3O^+ forms phenol and a by product **P**. Reaction of **P** with 3 equivalents of Cl_2 gives compound **Q**. Treatment of **Q** with $Ca(OH)_2$ produces compound **R** and calcium salt **S**.
The correct statement(s) regarding **P**, **Q**, **R** and **S** is(are)

(A) Reaction of **P** with **R** in the presence of KOH followed by acidification gives

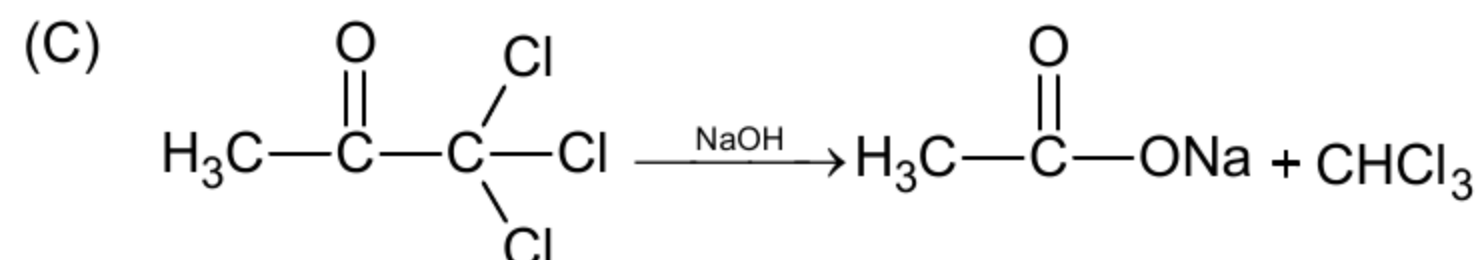
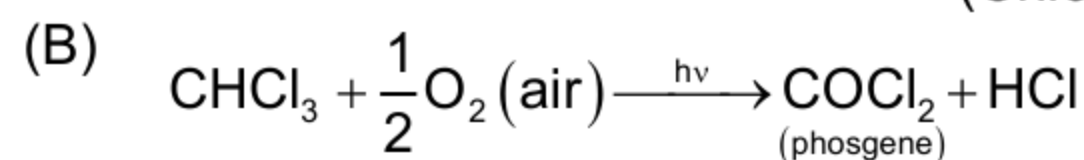
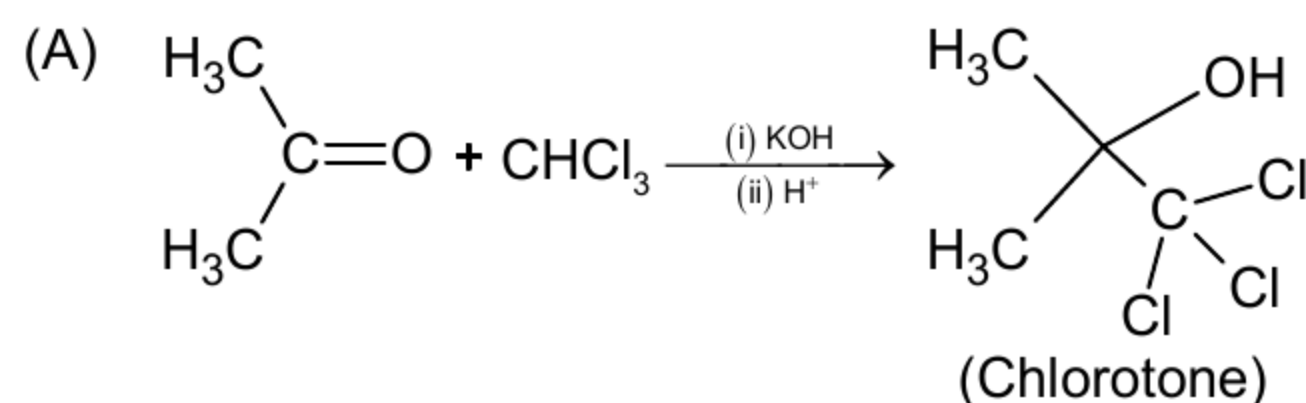
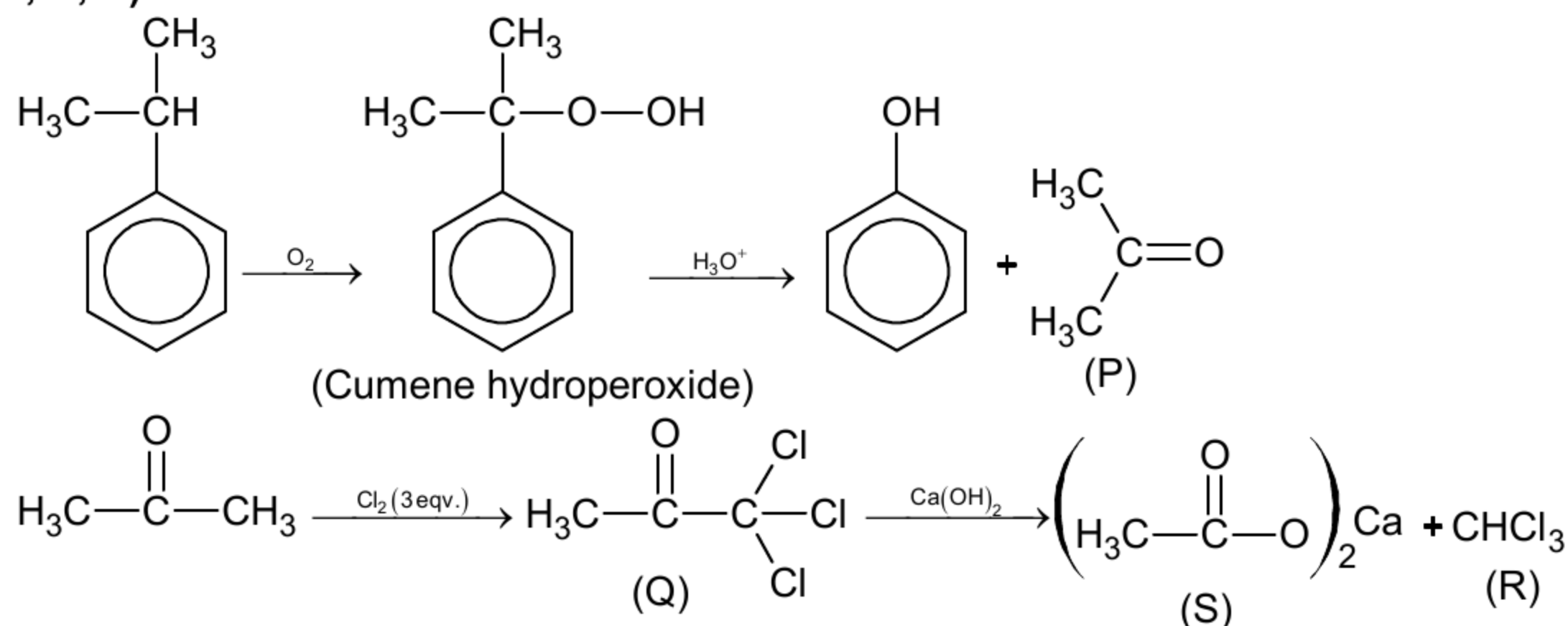


(B) Reaction of **R** with O_2 in the presence of light gives phosgene gas

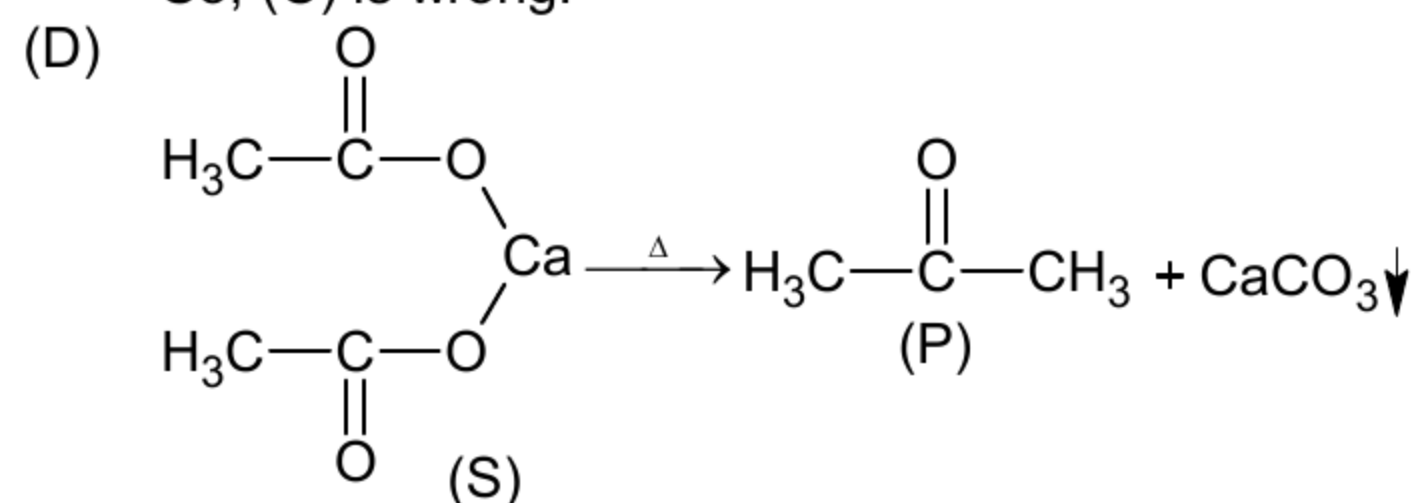
(C) **Q** reacts with aqueous NaOH to produce Cl_3CCH_2OH and $Cl_3CCOONa$

(D) **S** on heating gives **P**

Ans. Sol. (A, B, D)



So, (C) is wrong.



***Q7.** The option(s) in which at least three molecules follow Octet Rule is(are)

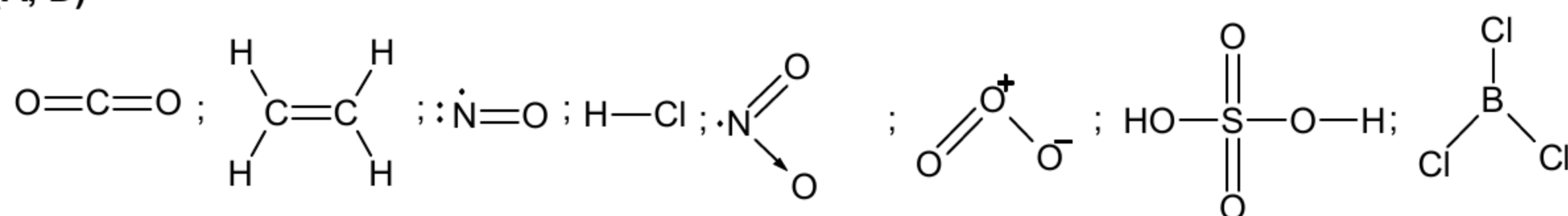
(A) CO_2 , C_2H_4 , NO and HCl

(B) NO_2 , O_3 , HCl and H_2SO_4

(C) BCl_3 , NO, NO_2 and H_2SO_4

(D) CO_2 , BCl_3 , O_3 and C_2H_4

Ans. Sol. (A, D)

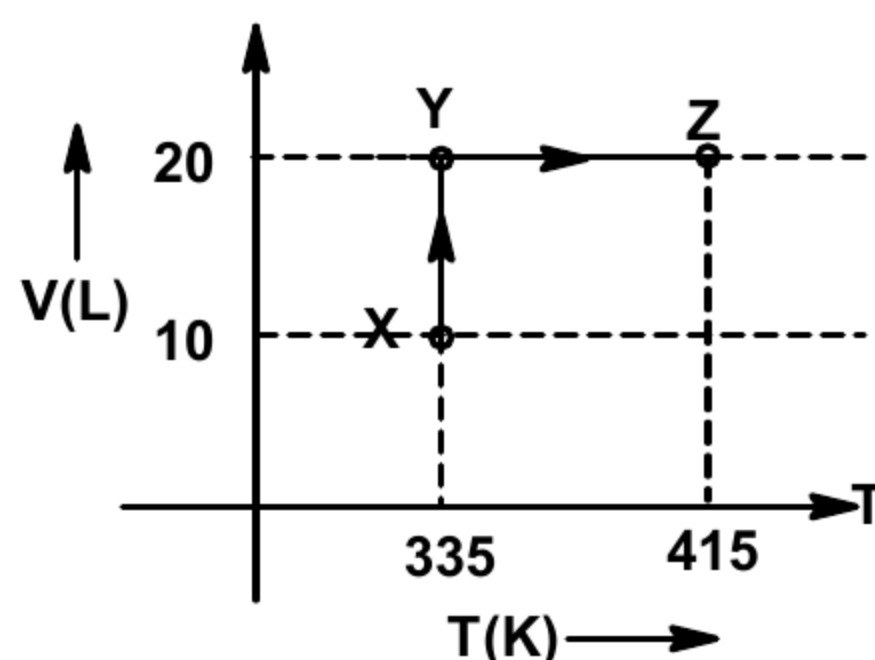


Only CO_2 , C_2H_4 , H-Cl and O_3 follows octet rule.

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct integer is entered;
Zero Marks : 0 In all other cases.

***Q8.** Consider the following volume – temperature ($V - T$) diagram for the expansion of 5 moles of an ideal monoatomic gas.



Consider only $P - V$ work is involved, the total change in enthalpy (in Joule) for the transformation of state in the sequence $X \rightarrow Y \rightarrow Z$ is _____.

[Use the given data: Molar heat capacity of the gas for the given temperature range, $C_{v,m} = 12 \text{ J K}^{-1} \text{ mol}^{-1}$ and gas constant, $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$]

Ans. (8120)

Sol. Since $X \rightarrow Y$ is an isothermal process ($dT = 0$).

$$\text{So, } \Delta H_{X \rightarrow Y} = nC_{p,m}dT = 0$$

$Y \rightarrow Z$ is isochoric.

$$\Delta U_{Y \rightarrow Z} = nC_{v,m}dT = 5 \times 12 \times (415 - 335)$$

$$\Delta U_{Y \rightarrow Z} = 4800 \text{ J}$$

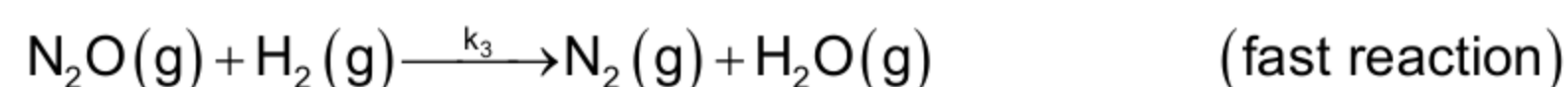
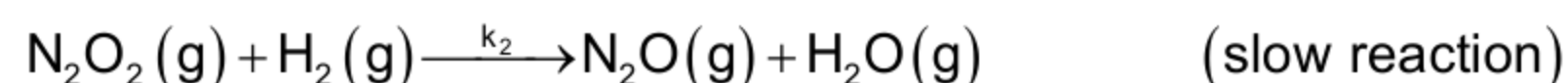
$$\Delta H_{Y \rightarrow Z} = \Delta U_{Y \rightarrow Z} + \Delta(PV)$$

$$= \Delta U_{Y \rightarrow Z} + nR\Delta T = 4800 + 5 \times 8.3 \times 80 = 8120 \text{ J}$$

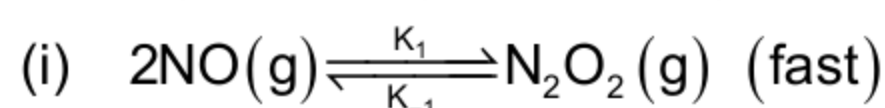
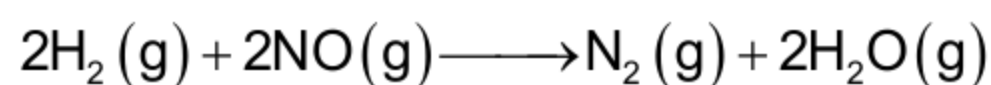
***Q.9** Consider the following reaction,



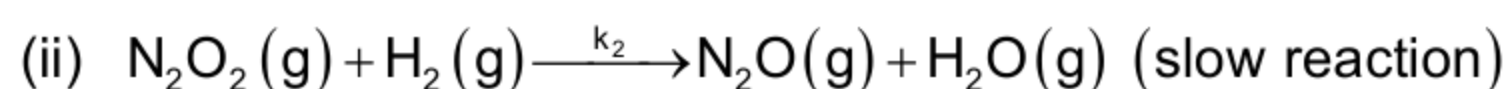
Which follows the mechanism given below?



The order of the reaction is _____.

Ans. (3)**Sol.** Net reaction:

$$\frac{k_1}{k_{-1}} = \frac{[\text{N}_2\text{O}_2]}{[\text{NO}]^2} \quad \dots(1)$$



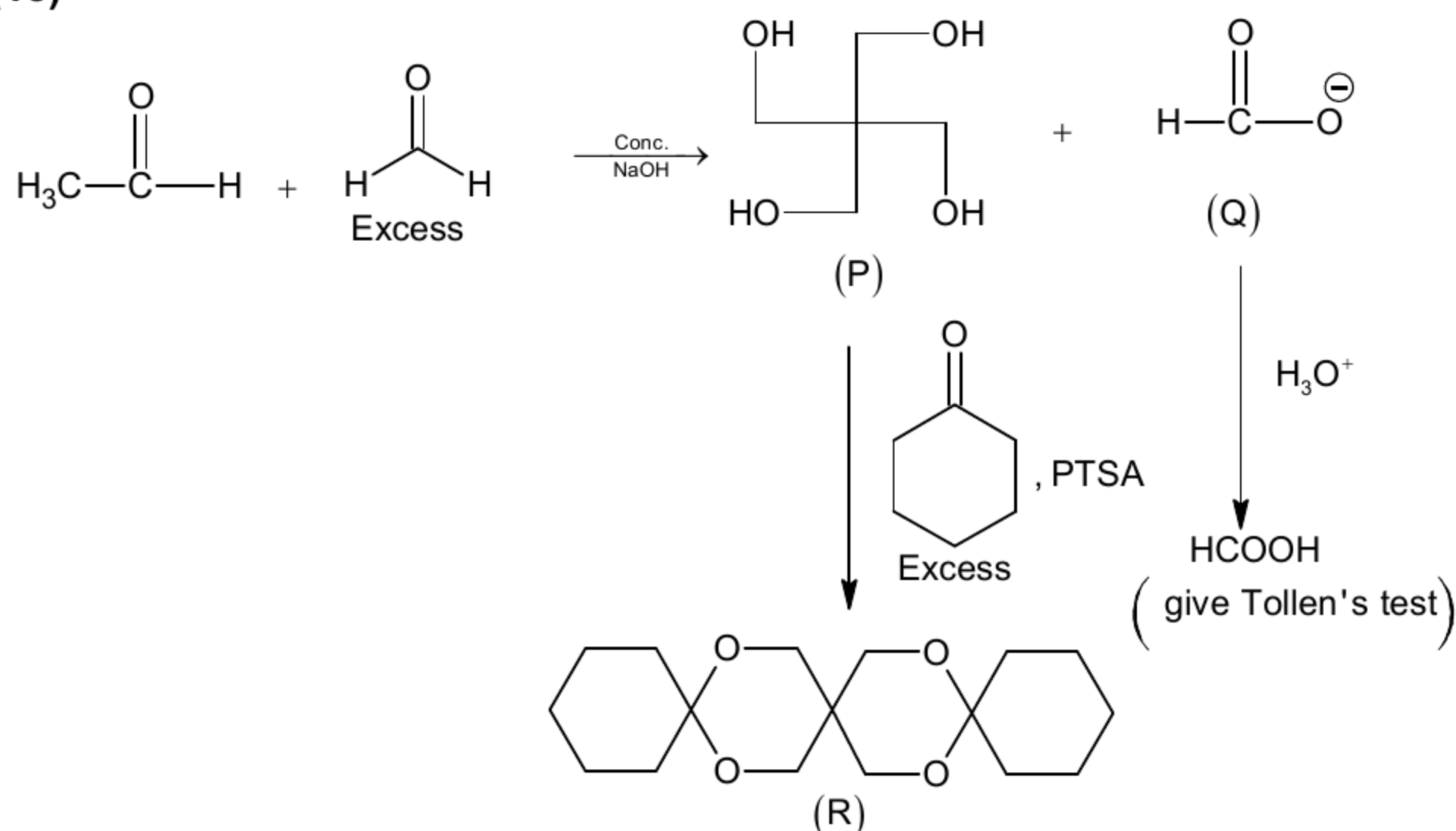
$$r = k_2 [\text{N}_2\text{O}_2][\text{H}_2]$$

$$= k_2 \times \frac{k_1}{k_{-1}} [\text{NO}]^2 \times [\text{H}_2]$$

$$r = \frac{k_2 k_1}{k_{-1}} [\text{NO}]^2 [\text{H}_2]$$

So, order of reaction = 3.

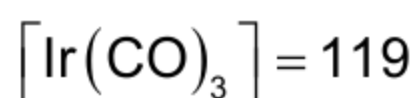
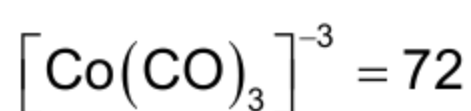
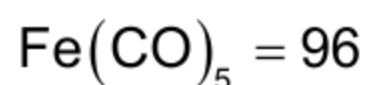
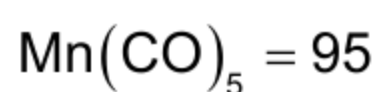
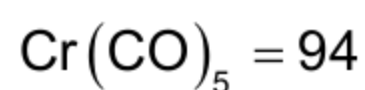
Q.10 Complete reaction of acetaldehyde with excess formaldehyde, upon heating with conc. NaOH solution, gives **P** and **Q**. Compound **P** does not give Tollen's test, whereas **Q** on acidification gives positive Tollen's test. Treatment of **P** with excess cyclohexanone in the presence of catalytic amount of p-toluenesulfonic acid (PTSA) gives product **R**. Sum of the number of methylene groups ($-\text{CH}_2-$) and oxygen atoms in **R** is _____.

Ans. (18)**Sol.**The number of CH_2 group in **R** = 14The number of O atom in **R** = 4

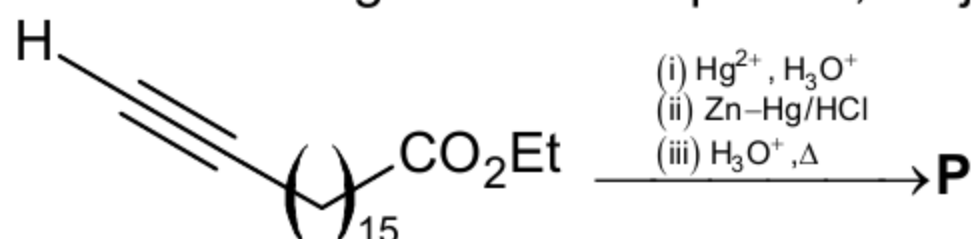
Total = 18

Q.11 Among $\text{V}(\text{CO})_6$, $\text{Cr}(\text{CO})_5$, $\text{Mn}(\text{CO})_5$, $\text{Fe}(\text{CO})_5$, $[\text{Co}(\text{CO})_3]^{3-}$, $[\text{Cr}(\text{CO})_4]^{4-}$ and $\text{Ir}(\text{CO})_3$, the total number of species isoelectronic with $\text{Ni}(\text{CO})_4$ is _____.
[Given, atomic number: V = 23, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Ir = 77]

Ans. (1)**Sol.** $\text{Ni}(\text{CO})_4 = 84$ $\text{V}(\text{CO})_6 = 107$



Q.12 In the following reaction sequence, major product **P** is formed.



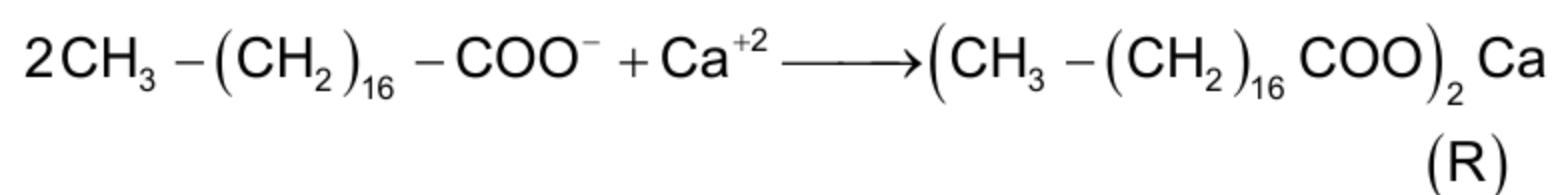
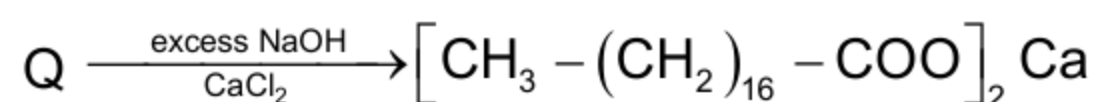
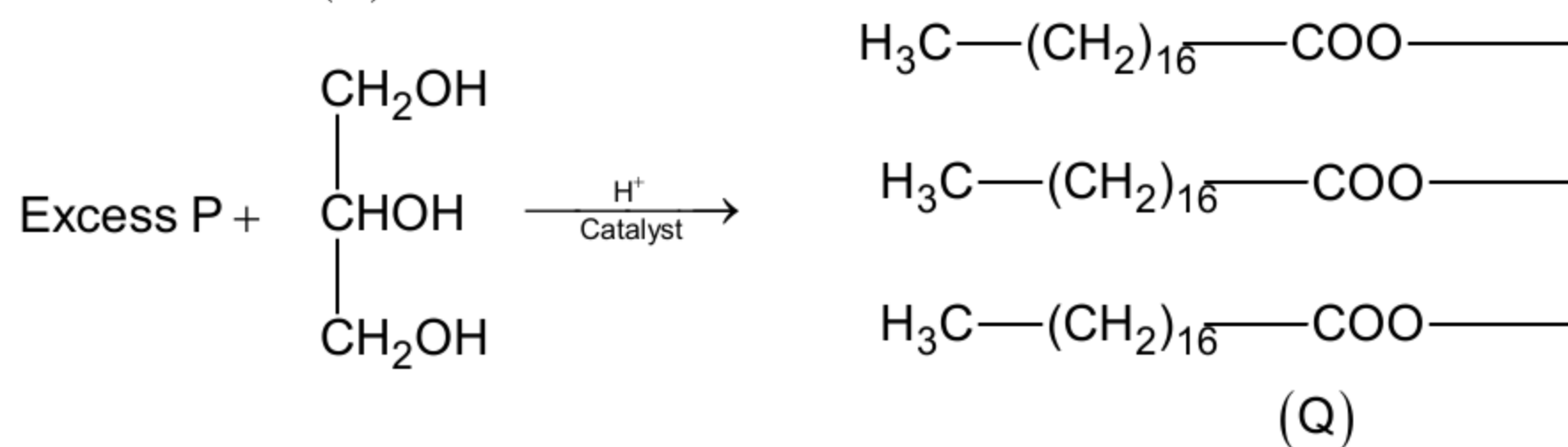
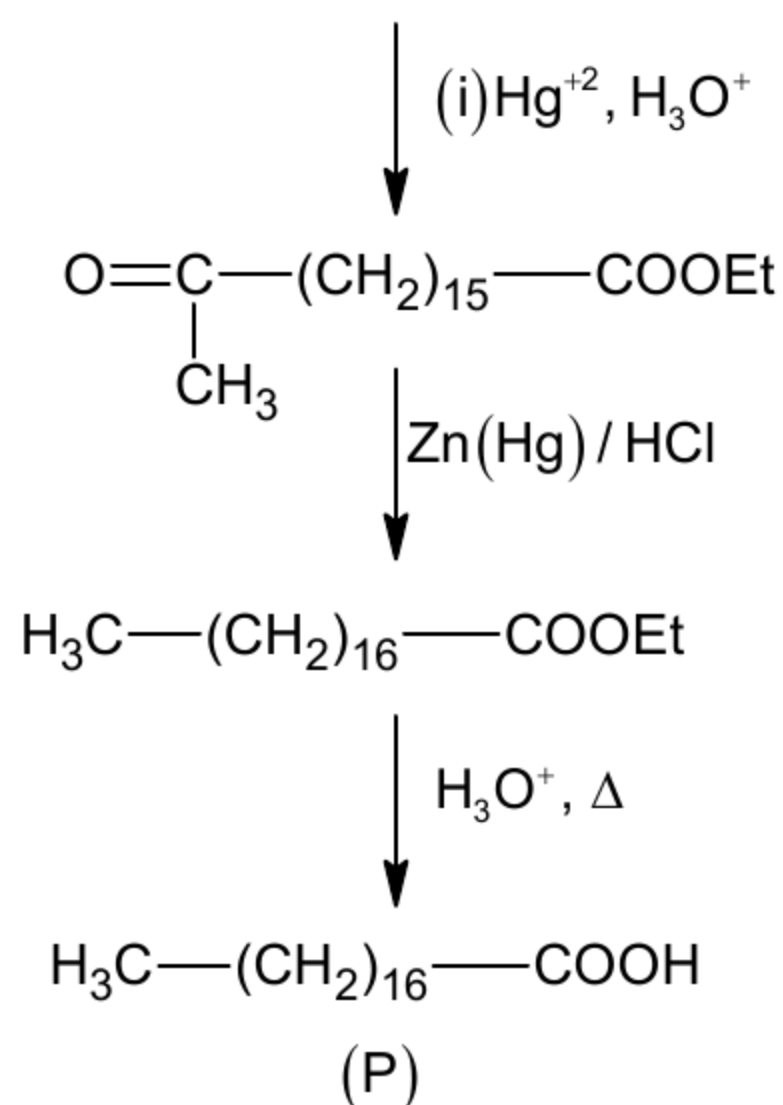
Glycerol reacts completely with excess **P** in the presence of an acid catalyst to form **Q**. Reaction of **Q** with excess NaOH followed by the treatment with CaCl_2 yields Ca – soap **R**, quantitatively.

Starting with one mole of **Q**, the amount of **R** produced in gram is _____.

[Given, atomic weight: H = 1, C = 12, N = 14, O = 16, Na = 23, Cl = 35, Ca = 40]

Ans. (909)

Sol. $\text{H}-\text{C}\equiv\text{C}-(\text{CH}_2)_{15}-\text{COOEt}$



Mole of R = 1.5
 Wt. of R = 1.5×606
 Wt. of R = 909

- Q. 13** Among the following complexes, the total number of diamagnetic species is _____.
 $[\text{Mn}(\text{NH}_3)_6]^{3+}$, $[\text{MnCl}_6]^{3-}$, $[\text{FeF}_6]^{3-}$, $[\text{CoF}_6]^{3-}$, $[\text{Fe}(\text{NH}_3)_6]^{3+}$ and $[\text{Co}(\text{en})_3]^{3+}$
 [Given, atomic number: Mn = 25, Fe = 26, Co = 27; en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$]

Ans. (1)

- Sol.**
1. $[\text{Mn}(\text{NH}_3)_6]^{3+}$, $n = 2$, Paramagnetic
 2. $[\text{MnCl}_6]^{3-}$, $n = 4$, Paramagnetic
 3. $[\text{FeF}_6]^{3-}$, $n = 5$, Paramagnetic
 4. $[\text{CoF}_6]^{3-}$, $n = 4$, Paramagnetic
 5. $[\text{Fe}(\text{NH}_3)_6]^{3+}$, $n = 1$, Paramagnetic
 6. $[\text{Co}(\text{en})_3]^{3+}$, $n = 0$, Diamagnetic

SECTION 4 (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
 Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
 Negative Marks : -1 In all other cases.

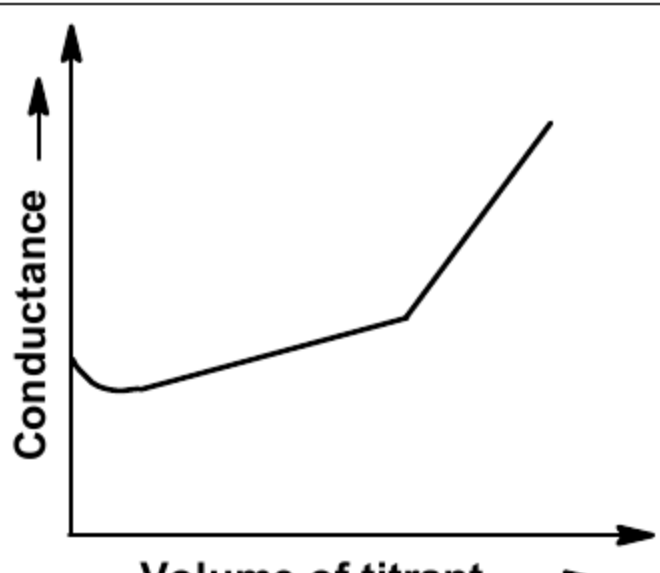
- Q.14** In a conductometric titration, small volume of titrant of higher concentration is added stepwise to a larger volume of titrate of much lower concentration, and the conductance is measured after each addition.

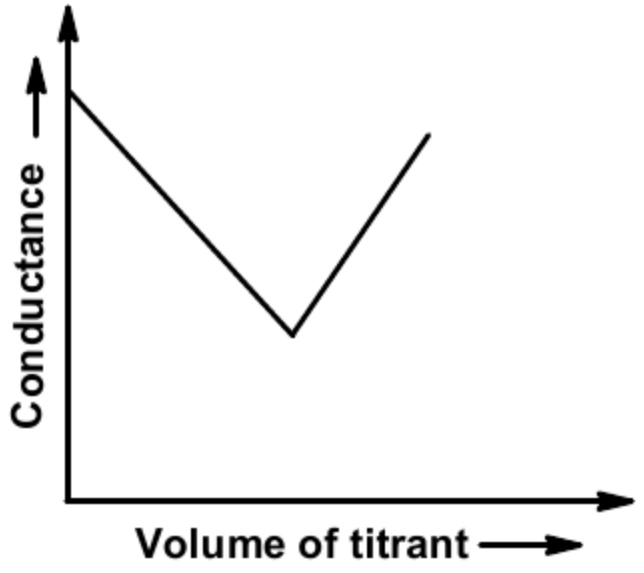
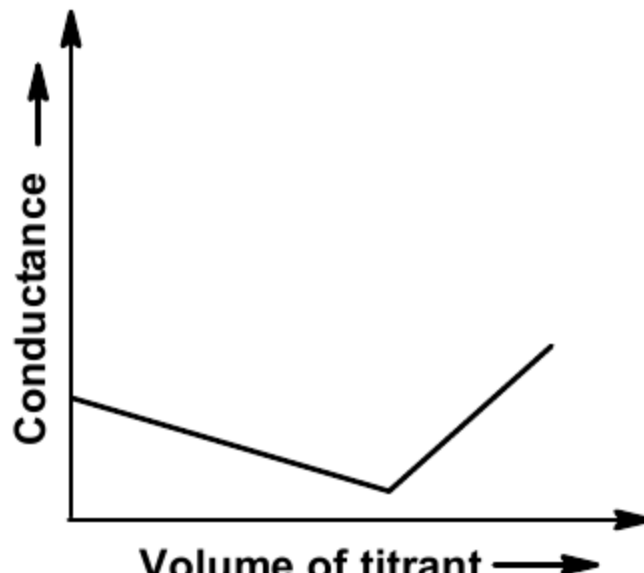
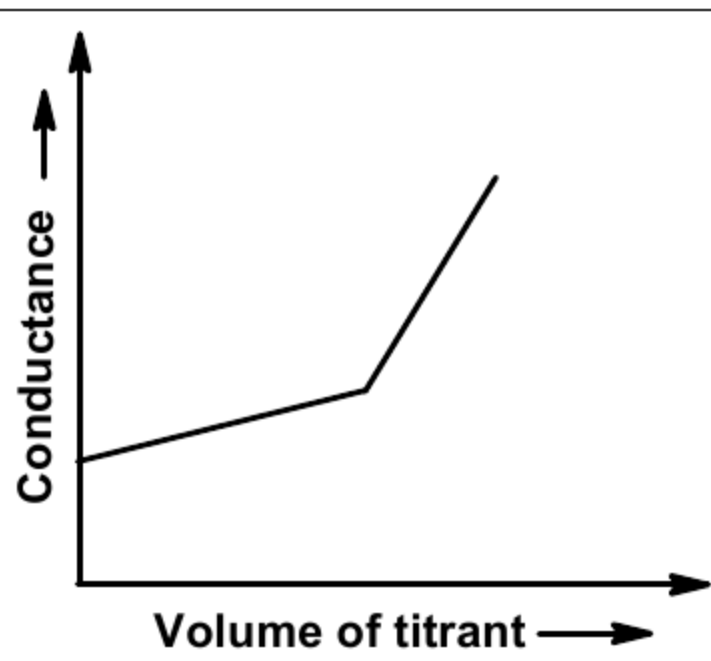
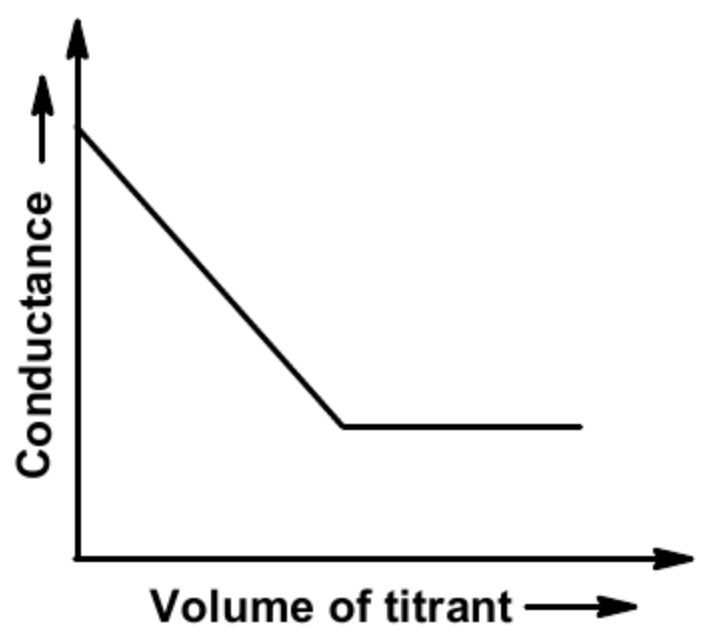
The limiting ionic conductivity (Λ_0) values (in $\text{mS m}^2 \text{mol}^{-1}$) for different ions in aqueous solutions are given below:

Ions	Ag^+	K^+	Na^+	H^+	NO_3^-	Cl^-	SO_4^{2-}	OH^-	CH_3COO^-
(Λ_0)	6.2	7.4	5.0	35.0	7.2	7.6	16.0	19.9	4.1

For different combinations of titrates and titrants given in **List-I**, the graphs of 'conductance' versus 'volume of titrant' are given in **List-II**.

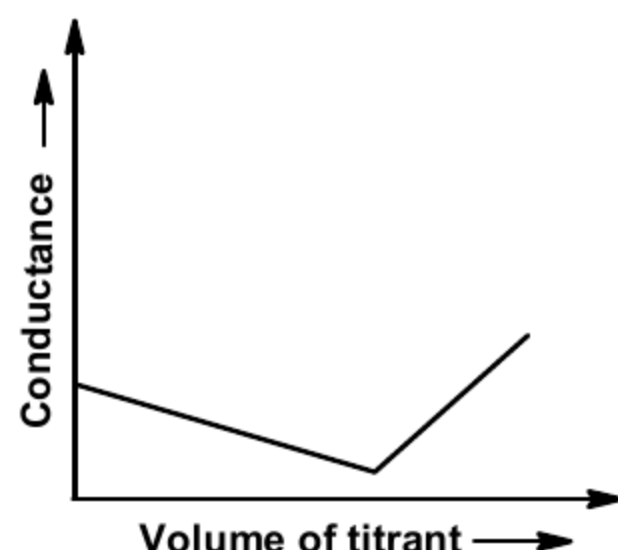
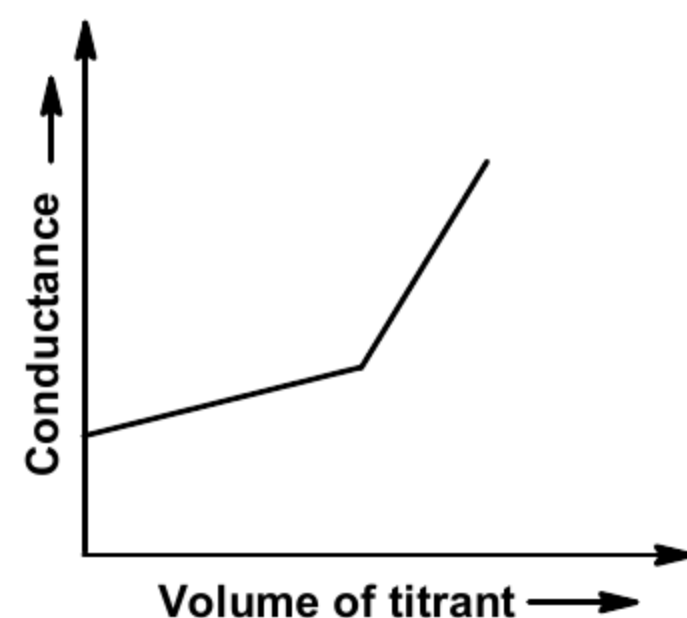
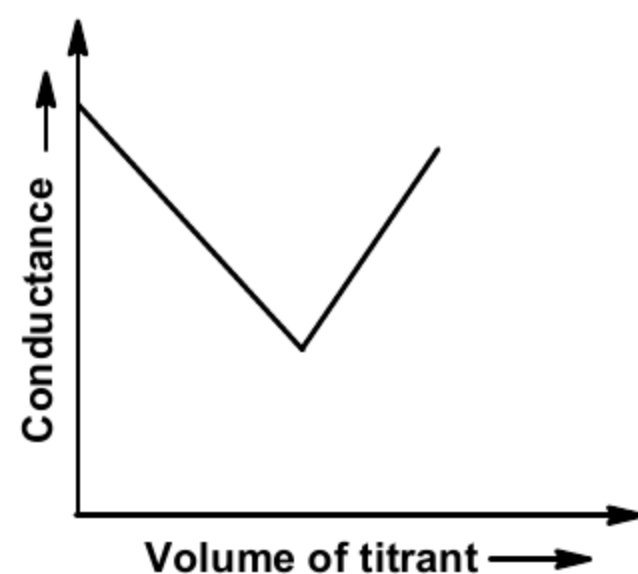
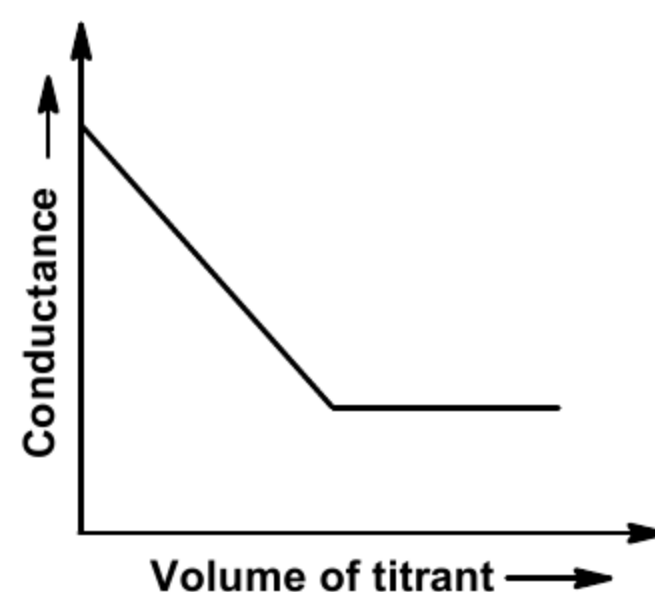
Match each entry **List-I** with the appropriate entry in **List-II** and choose the correct option.

List-I		List-II	
(P)	Titrate: KCl Titrant: AgNO_3	(1)	

(Q)	Titrate: AgNO_3 Titrant: KCl	(2)	
(R)	Titrate: NaOH Titrant: HCl	(3)	
(S)	Titrate: NaOH Titrant: CH_3COOH	(4)	
		(5)	

- (A) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (2), (S) $\rightarrow$ (5)
 (B) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (4), (R) $\rightarrow$ (3), (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (4), (R) $\rightarrow$ (2), (S) $\rightarrow$ (5)
 (D) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (2), (S) $\rightarrow$ (1)

Ans. (C)

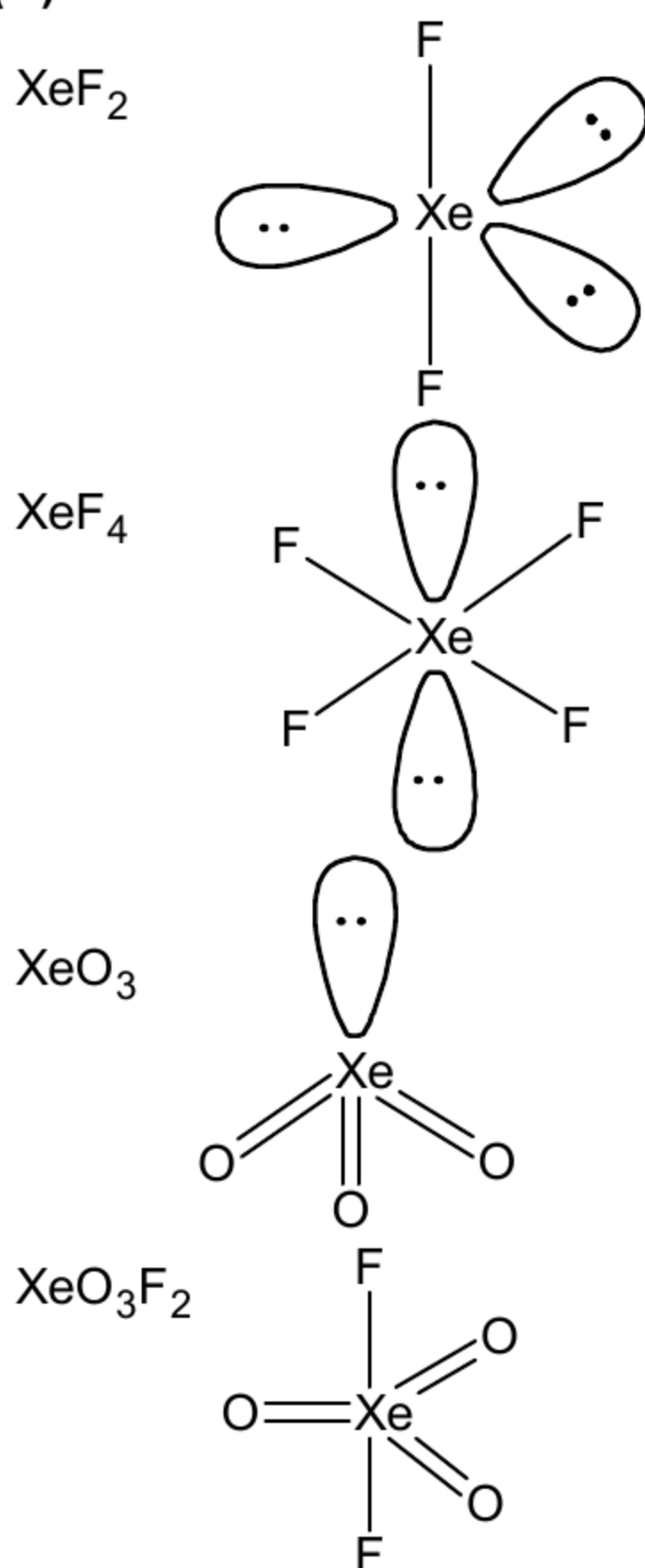
Sol. (P) $\rightarrow$ (3)(Q) $\rightarrow$ (4)(R) $\rightarrow$ (2)(S) $\rightarrow$ (5)

***Q.15** Based on **VSEPR** model, match the xenon compounds given in **List-I** with the corresponding geometries and the number of lone pairs on xenon given in **List-II** and choose the correct option.

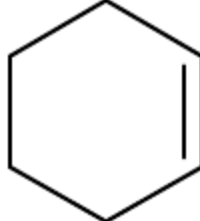
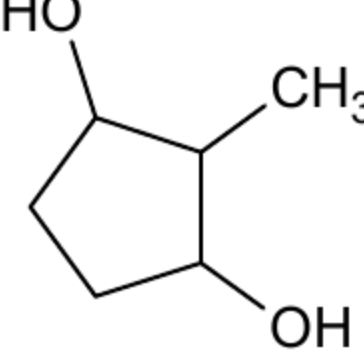
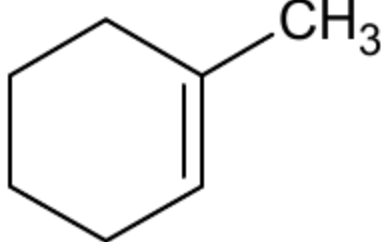
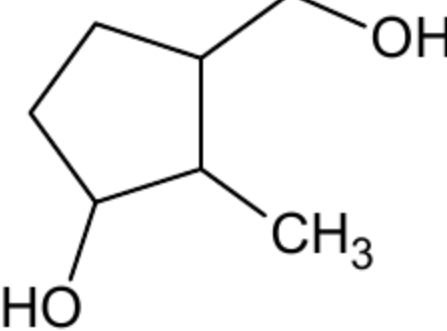
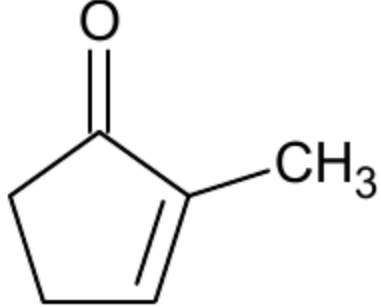
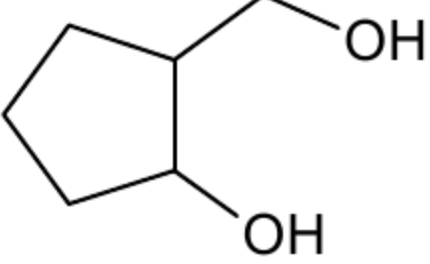
List-I		List-II	
(P)	XeF_2	(1)	Trigonal bipyramidal and two lone pair of electrons
(Q)	XeF_4	(2)	Tetrahedral and one lone pair of electrons
(R)	XeO_3	(3)	Octahedral and two lone pair of electrons
(S)	XeO_3F_2	(4)	Trigonal bipyramidal and no lone pair of electrons
		(5)	Trigonal bipyramidal and three lone pair of electrons

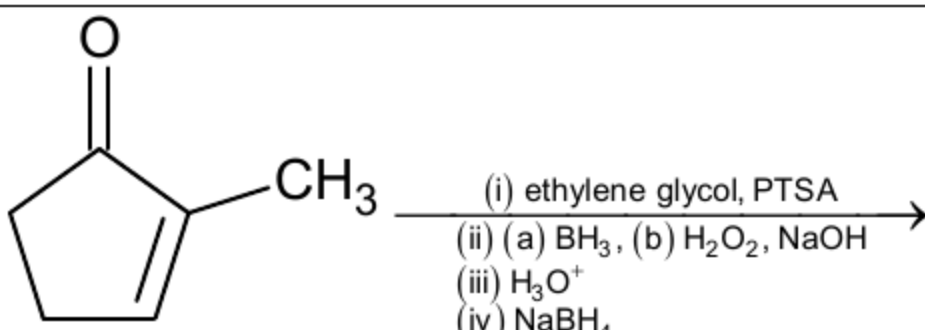
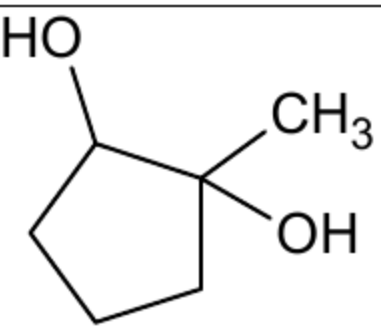
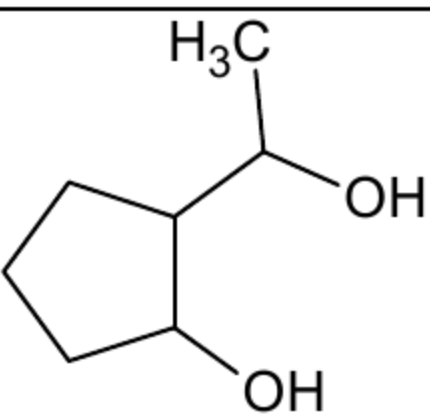
- (A) (P) $\rightarrow$ (5), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (1)
 (B) (P) $\rightarrow$ (5), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (2), (S) $\rightarrow$ (4)
 (C) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (2), (S) $\rightarrow$ (1)
 (D) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (5), (S) $\rightarrow$ (3)

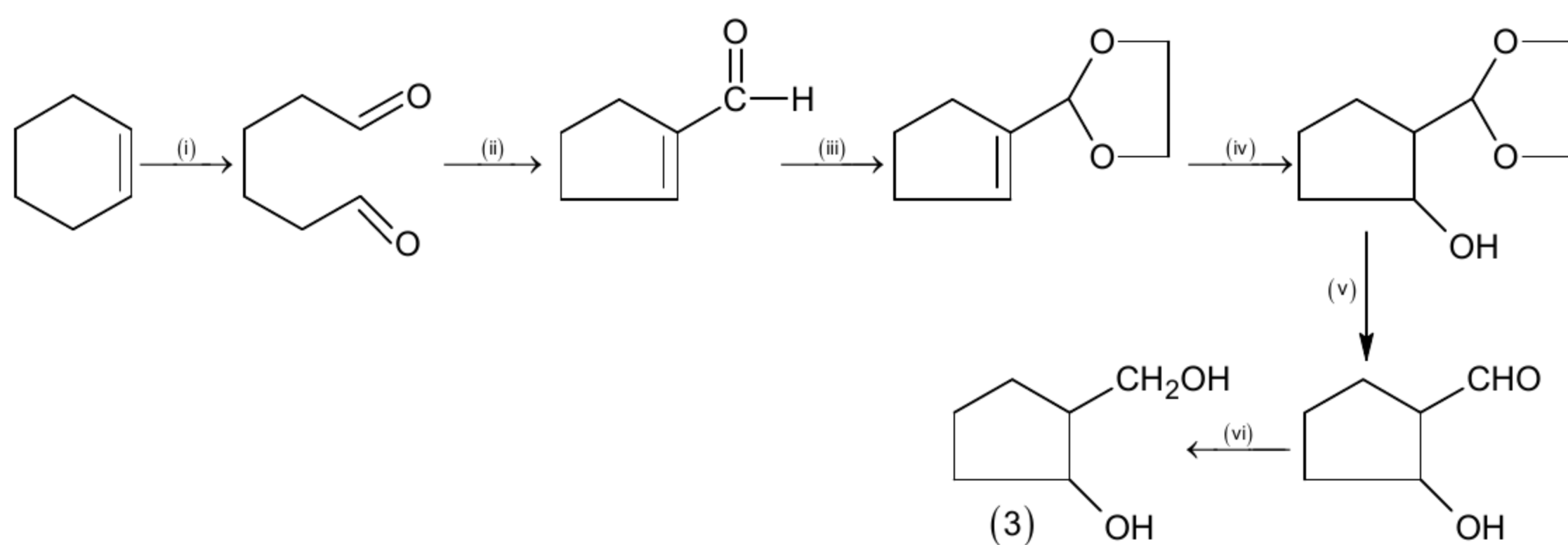
Ans. (B)
Sol.



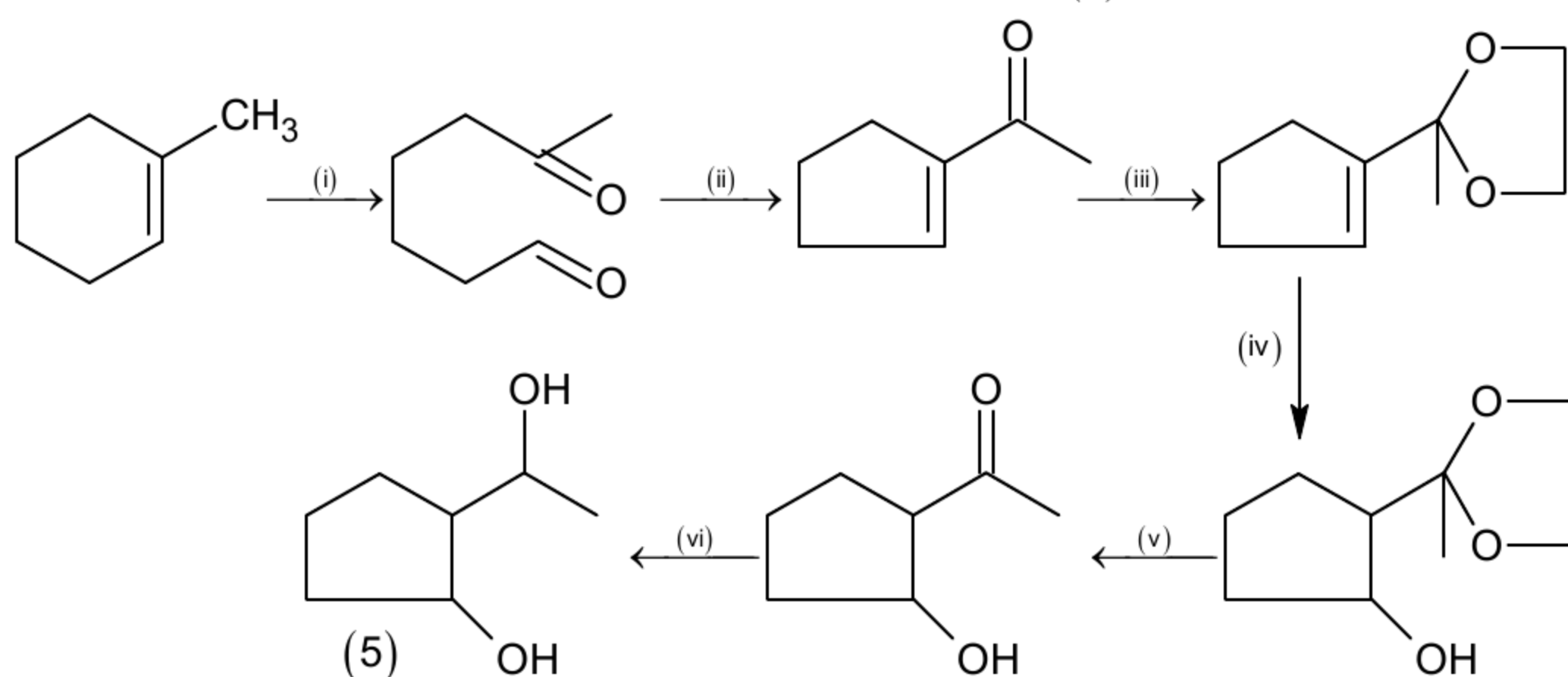
Q.16 **List-I** contains various reaction sequence and **List-II** contains the possible products. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

List-I		List-II	
(P)	 $\xrightarrow{\begin{matrix} \text{(i) O}_3, \text{Zn} \\ \text{(ii) aq. NaOH}, \Delta \\ \text{(iii) ethylene glycol, PTSA} \\ \text{(iv) (a) BH}_3, \text{(b) H}_2\text{O}_2, \text{NaOH} \\ \text{(v) H}_3\text{O}^+ \\ \text{(vi) NaBH}_4 \end{matrix}}$	(1)	
(Q)	 $\xrightarrow{\begin{matrix} \text{(i) O}_3, \text{Zn} \\ \text{(ii) aq. NaOH}, \Delta \\ \text{(iii) ethylene glycol, PTSA} \\ \text{(iv) (a) BH}_3, \text{(b) H}_2\text{O}_2, \text{NaOH} \\ \text{(v) H}_3\text{O}^+ \\ \text{(vi) NaBH}_4 \end{matrix}}$	(2)	
(R)	 $\xrightarrow{\begin{matrix} \text{(i) ethylene glycol, PTSA} \\ \text{(ii) (a) Hg(OAc)}_2, \text{H}_2\text{O}, \text{(b) NaBH}_4 \\ \text{(iii) H}_3\text{O}^+ \\ \text{(iv) NaBH}_4 \end{matrix}}$	(3)	

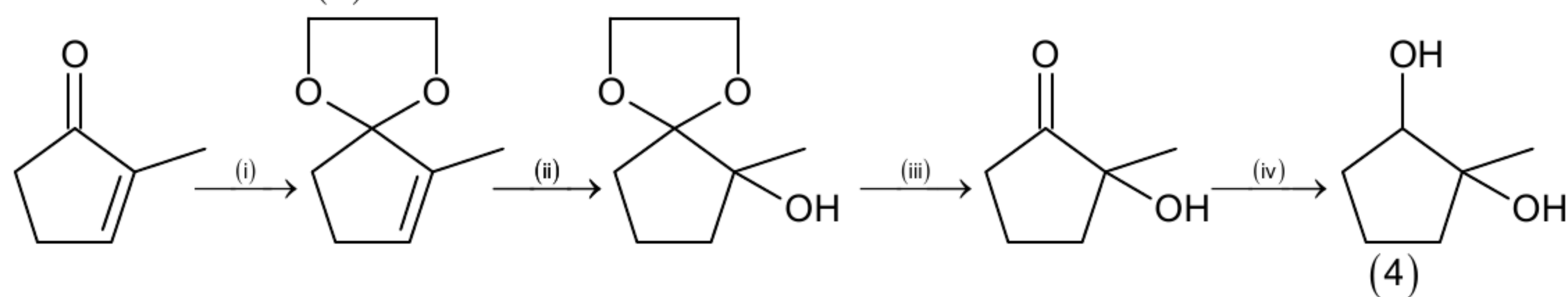
(S)		(4)	
		(5)	

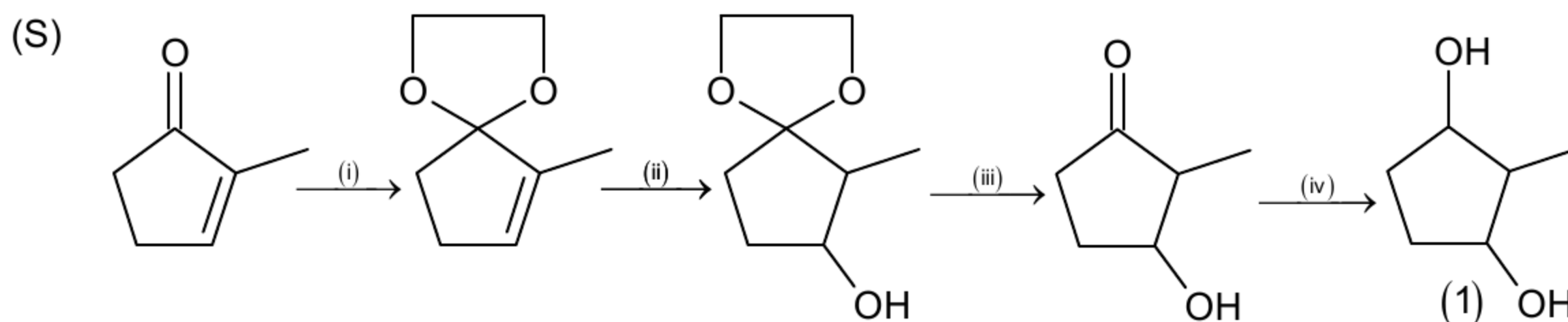
(A) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (5), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)(B) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)(C) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (5), (R) $\rightarrow$ (1), (S) $\rightarrow$ (4)(D) (P) $\rightarrow$ (5), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)**Ans.
Sol.****(A)**
(P)

(Q)



(R)





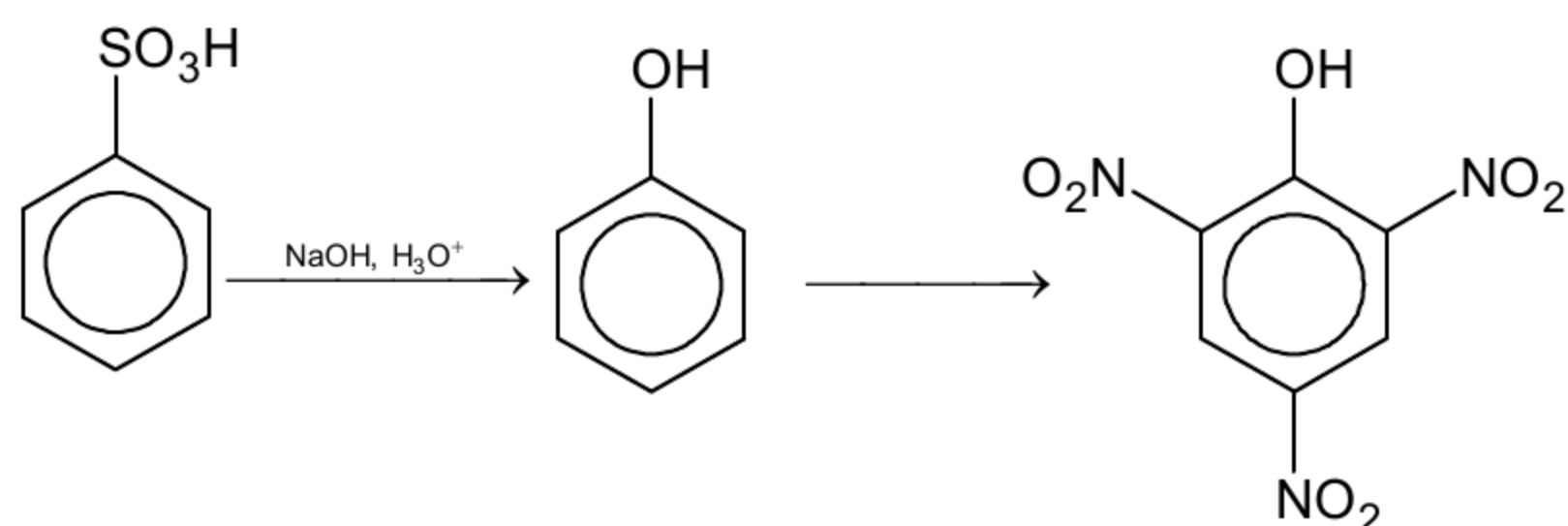
Q.17 **List-I** contains various reaction sequence and **List-II** contains different phenolic compounds. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

List-I		List-II	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	
		(5)	

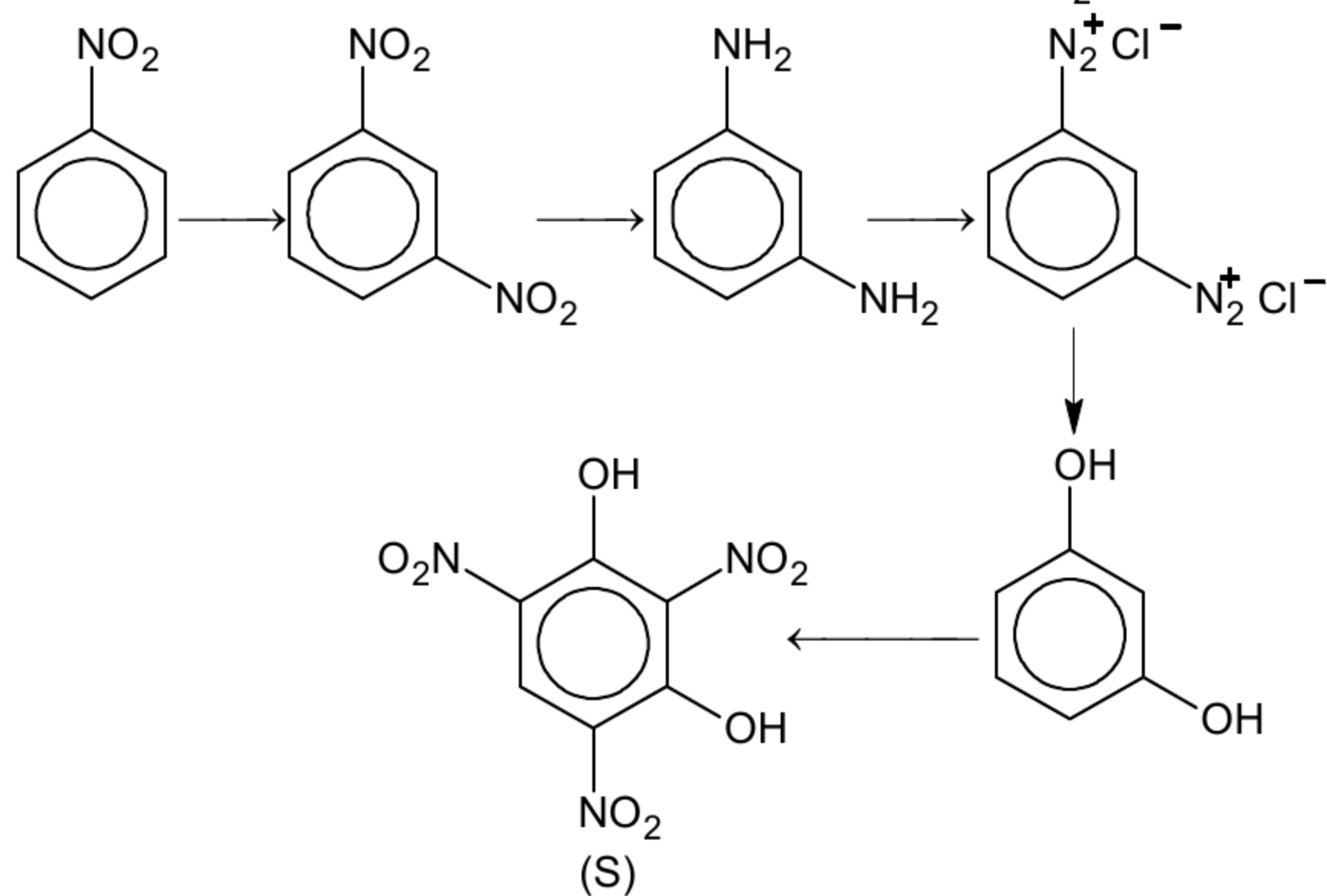
- (A) (P) → (2), (Q) → (3), (R) → (4), (S) → (5)
 (B) (P) → (2), (Q) → (3), (R) → (5), (S) → (1)
 (C) (P) → (3), (Q) → (5), (R) → (4), (S) → (1)
 (D) (P) → (3), (Q) → (2), (R) → (5), (S) → (4)

Ans. (C)

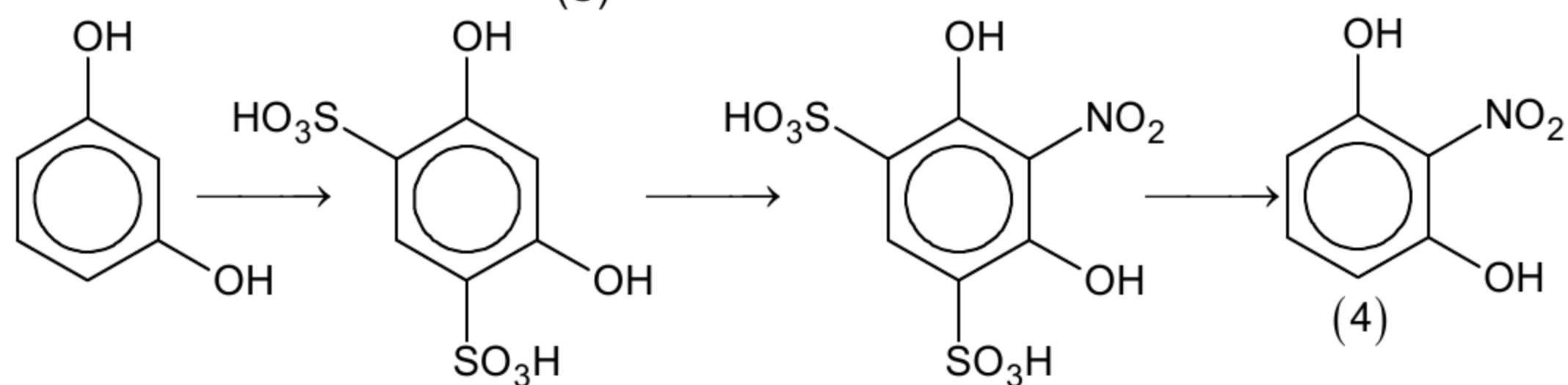
Sol. (P)



(Q)



(R)



(S)

